MET CS662 - Assignment #3

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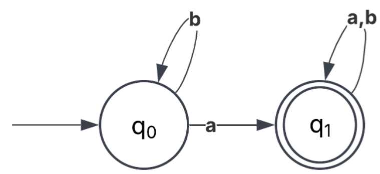
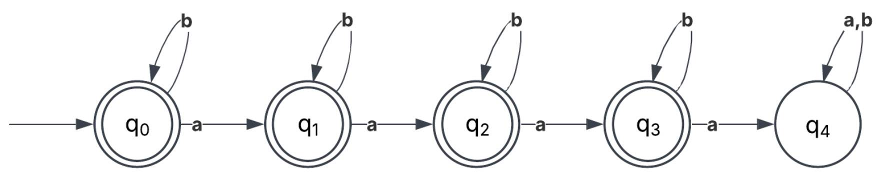
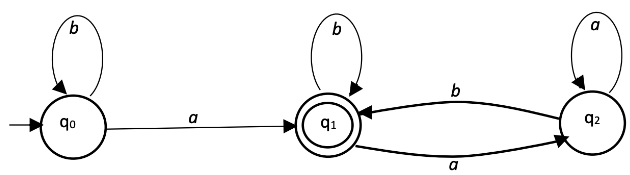
1. For $\sum = \{a, b\}$, construct deterministic finite automata that accept the sets consisting of
   a. All strings with at least one a.



   b. All strings with no more than three a's.



2. Give a set notation description of the language accepted by the automaton depicted in the following diagram. Can you think of a simple verbal characterization of the language?

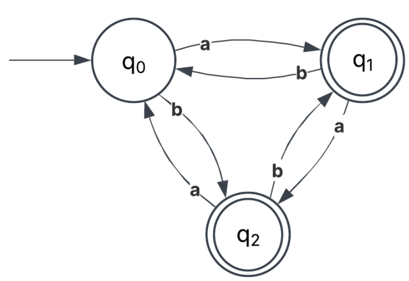


$$L = \{b^n a | n \geq 0\} \cup \{wawb | w \in \{a, b\}^*\}$$

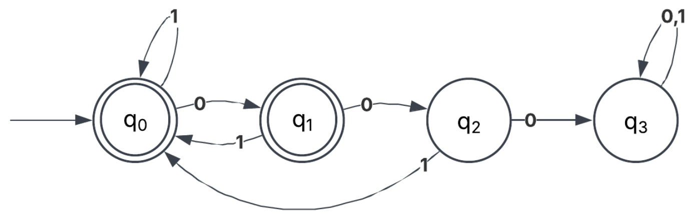
This language contains strings where there is exactly one a if the string ends with a or there is at least one a if the string ends with b.

3. Find a deterministic finite automaton for the following language on $\sum = \{a, b\}$

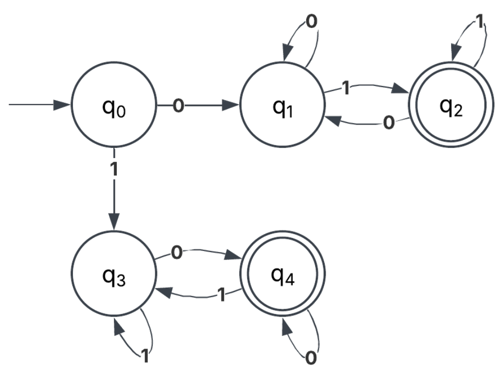
$$L = \{\omega \mid (n_a(\omega) - n_b(\omega)) \ mod\ 3 > 0 \}$$



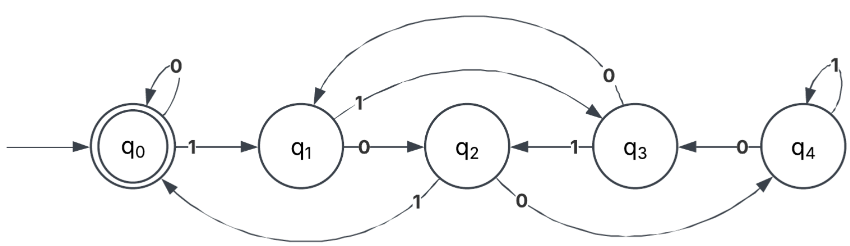
4. Consider the number of strings on $\{0, 1\}$ defined by the requirement below. For each construct a dfa.
    a. Every 00 is followed by a 1. For example, the strings 101, 0010, 0010011001 are in the language, but 0001 and 00100 are not.



    b. The leftmost symbol differs from the rightmost one.



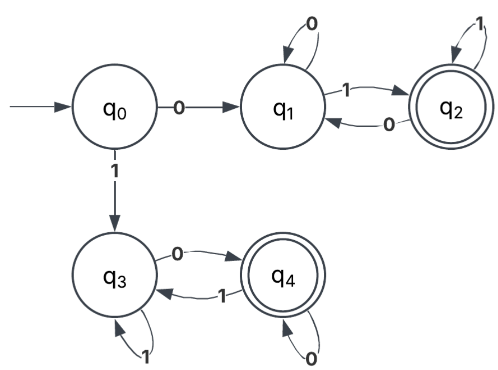
5. Construct a dfa that accepts strings on $\{0, 1\}$ if and only if the value of the string, interpreted as a binary representation of an integer, is zero modulo five. For example, 0101 and 1111, representing the integers 5 and 15, respectively, are to be accepted.



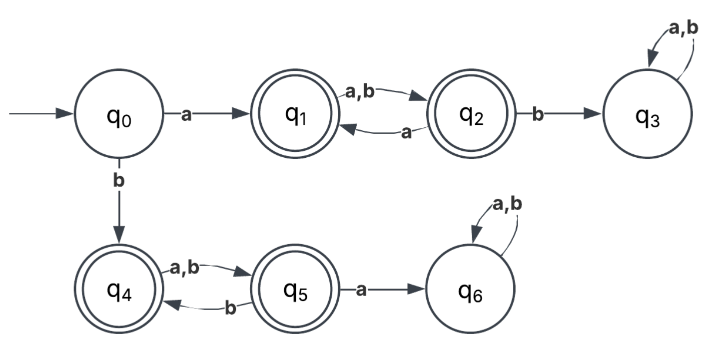
6. Show that if L is regular, so is $L - \{\lambda\}$.
   Suppose that an automaton M exists that recognizes L. One must exist because L is assumed to be a regular language.
   $$M = (Q, \Sigma, \delta, q_0, F)$$
   Where Q is the set of states, Σ is the alphabet, δ is the transition function, q_0 is the start state, and F is the set of accepting states. Set $L' = L - \{\lambda\}$.
   If $\delta(q_0, \lambda) \notin F$, then $\lambda \notin L$. So $L' = L$, and L is regular. Therefore, $L - \{\lambda\}$ is also regular.
   If $\delta(q_0, \lambda) \in F$, then $\lambda \in L$. So $L' \neq L$. Construct a new automaton
   $$M' = (Q', \Sigma, \delta', q_0', F)$$
   where $Q' = Q \cup \{q_0'\}$ and $\delta' = \delta \cup \left\{\delta'^*(q_0', w) = \delta^*(q_0, w) \middle| \forall w \in \Sigma^+\right\} \cup \{\delta'(q_0', \lambda) = q_0'\}$. The automaton A' has been constructed so that $\delta'(q_0', w) \in F, \forall w \in \Sigma^+$, but $\delta'(q_0', \lambda) \notin F$. Therefore, M' recognizes L', which proves that $L - \{\lambda\}$ is a regular language.
   This proof relies upon constructing a modified automaton M', that is the same as M except that there is a new start state $q_0' \notin F$. The transition function is modified to maintain the original transitions but add in transitions to ensure that $q_0' \times \Sigma \to Q = q_0 \times \Sigma \to Q$ for any symbol other than λ, adding that $q_0' \times \lambda \to q_0' \notin F$. Therefore, all the words in L are still recognized by M', except for λ.

For the next few questions, design a DFA for the defined languages…

7. $L = \{w \in \Sigma^* |\ w$ does not contain an equal number of occurrences of the substrings 01 and 10$\}$ with $\Sigma = \{0, 1\}$. Examples of accepted strings: 10, 01, 1010, 1110110, 000011101, etc. notice for example in 1010 there 2 substrings 10 and a single substring 01.



8. The set of all strings whose 1st, 3rd, 5th, … characters are the same, $\sum = \{a, b\}$. Examples of accepted strings: a, a, aa, ab, aba, bbb, bb, ba, ababab, bbbbbbbbb, bbbabbbab…etc.



9. Set of all strings that are at least of length 4 and contains even number of 1's. $\sum = \{0, 1\}$. For example: 1010, 11011, 0000, 11100111, …etc.